Anesthesia for Orthopedics MCQ – Chapter 36

Source: Short Textbook of Anesthesia – Ajay Yadav

Section 8: Subspecialty Anesthetic Management (Chapter 36)

50 Questions • 30 Minutes

Instructions:
- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

Questions

Q1. What special problems does orthopedic anesthesia pose? **A.** Only blood loss **B.** Only tourniquet problems **C.** Lengthy procedures, increased blood loss, positioning, tourniquet-related problems, fat embolism and DVT **D.** Only positioning issues	**Ans: C**
Q2. What complications are associated with tourniquet during inflation? **A.** Hypothermia **B.** Hypertension, increased cardiac output and increased pulmonary artery pressure **C.** Hypotension and decreased cardiac output **D.** Bradycardia	**Ans: B**
Q3. What metabolic change occurs within 6 minutes of tourniquet inflation? **A.** Local tissue ischemia leads to acidosis (mitochondrial pO_2 becomes zero) **B.** No metabolic change **C.** Alkalosis **D.** Hyperglycemia	**Ans: A**
Q4. How is pain during tourniquet inflation relieved? **A.** Only by deflation (not relieved by IV analgesics) **B.** By local anesthetics **C.** By IV analgesics **D.** By increasing tourniquet pressure	**Ans: A**
Q5. What causes hypotension during tourniquet deflation? **A.** Vasovagal reflex **B.** Sudden release of accumulated metabolites like thromboxane and substance P **C.** Blood loss **D.** Cardiac depression	**Ans: B**
Q6. Why does hypothermia occur during tourniquet deflation? **A.** Due to heat loss as acid metabolites are vasodilators **B.** Due to blood transfusion **C.** Due to air conditioning **D.** Due to cold fluids	**Ans: A**
Q7. What happens to end tidal CO_2 during tourniquet deflation? **A.** No change **B.** Decreases **C.** Becomes zero **D.** Transient increase (CO_2 accumulated in limb is released)	**Ans: D**

Q8. Which is the most crucial period during tourniquet use? **A.** Inflation **B.** Pre-inflation **C.** During surgery **D.** Deflation	**Ans: D**

Q9. What is the maximum duration for tourniquet pressure for upper and lower limbs? **A.** 4 hours for lower limbs **B.** Not more than 2 hours for both upper and lower limbs **C.** 3 hours **D.** 1 hour	**Ans: B**

Q10. If re-inflation of tourniquet is needed, how long should it be deflated first? **A.** 1 hour **B.** 10 minutes **C.** 5 minutes **D.** At least 30 minutes	**Ans: D**

Q11. What is the maximum tourniquet pressure above systolic? **A.** 150 mm Hg above systolic **B.** Not more than 100 mm Hg above systolic (maximum 250 mm Hg) **C.** 200 mm Hg above systolic **D.** 50 mm Hg above systolic	**Ans: B**

Q12. What complication may occur if tourniquet pressure is high or time is prolonged? **A.** Infection **B.** Thrombosis **C.** Neuropraxia **D.** Compartment syndrome	**Ans: C**

Q13. When is fat embolism syndrome (FES) usually seen? **A.** After cardiac surgery **B.** After spinal surgery **C.** After fracture of long bones and pelvis **D.** After abdominal surgery	**Ans: C**

Q14. When does fat embolism classically present after fracture? **A.** Immediately **B.** After 1 week **C.** Within 24-72 hours **D.** Within 6 hours	**Ans: C**

Q15. Is the incidence of FES higher in nailing or plating? **A.** Higher in plating **B.** Higher in nailing compared to plating **C.** Same in both **D.** Neither causes FES	**Ans: B**

Q16. What is the classic triad of fat embolism syndrome? **A.** Chest pain, cough, hemoptysis **B.** Headache, vomiting, seizures **C.** Dyspnea, confusion and petechial hemorrhages **D.** Fever, tachycardia, hypotension	**Ans: C**
Q17. What is pathognomonic of fat embolism syndrome? **A.** Confusion **B.** Dyspnea **C.** Tachycardia **D.** Petechial rash	**Ans: D**
Q18. What are the signs and symptoms of fat embolism? **A.** Only neurological symptoms **B.** Dyspnea, chest discomfort, tachycardia, hypotension, mental clouding, restlessness, seizures, delirium, petechial hemorrhage on upper chest, axilla and conjunctiva, hypoxia and finally coma and death **C.** Only petechial rash **D.** Only respiratory symptoms	**Ans: B**
Q19. What do fat globules in urine indicate regarding fat embolism? **A.** No significance **B.** Definitive diagnosis of FES **C.** Helpful in diagnosis but may not necessarily indicate progression to FES **D.** Rules out FES	**Ans: C**
Q20. What does X-ray chest show in fat embolism? **A.** Consolidation **B.** Pleural effusion **C.** Diffuse interstitial infiltrates **D.** Pneumothorax	**Ans: C**
Q21. What does capnography show in fat embolism? **A.** Increase in end tidal CO2 **B.** Fluctuating ETCO2 **C.** No change **D.** Fall in end tidal CO2 (ETCO2 may become zero if embolus is large enough to block main pulmonary artery)	**Ans: D**
Q22. What ECG finding is seen in fat embolism? **A.** Atrial fibrillation **B.** Right ventricular strain **C.** ST elevation **D.** Left ventricular strain	**Ans: B**

Q23. **What happens to serum lipase in fat embolism?** **A.** Increased (but has no relation to severity of disease) **B.** Absent **C.** Normal **D.** Decreased	**Ans: A**

Q24. **What is the first step in treatment of fat embolism?** **A.** Heparin therapy **B.** Ethanol infusion **C.** Early stabilization of fracture (may prevent fat embolism) **D.** Corticosteroids	**Ans: C**

Q25. **What ventilatory support is used if hypoxia persists on oxygen therapy in fat embolism?** **A.** CPAP only **B.** Non-invasive ventilation only **C.** High flow nasal cannula only **D.** Intermittent positive pressure ventilation (IPPV) with positive end expiratory pressure (PEEP)	**Ans: D**

Q26. **What is the pathophysiology of fat embolism in the lungs?** **A.** Causes pleural effusion **B.** Fat emboli cause endothelial injury which can precipitate ARDS and cerebral edema **C.** Causes bronchospasm **D.** Causes pneumothorax	**Ans: B**

Q27. **Are corticosteroids, heparin and ethanol considered effective treatments for fat embolism currently?** **A.** Only corticosteroids are effective **B.** No, they are now considered as outdated treatment modalities **C.** Only heparin is effective **D.** Yes, they are first-line treatment	**Ans: B**

Q28. **What is the incidence of fatal pulmonary embolism (PE) after hip surgery?** **A.** 5-10% **B.** Less than 0.5% **C.** 1-3% **D.** 10-15%	**Ans: C**

Q29. **In which orthopedic patients is the incidence of DVT and PE highest?** **A.** Upper limb surgery patients **B.** Arthroscopy patients **C.** Hand surgery patients **D.** Hip surgery patients particularly after prolonged immobilization	**Ans: D**

Q30. When should low molecular weight heparin be started for DVT prophylaxis in hip surgery? **A.** Immediately before surgery **B.** 24 hours before surgery **C.** Only postoperatively **D.** 12 hours or more in preoperative period and restart 12 hours after surgery	**Ans: D**
Q31. What serious complications have been reported in shoulder surgeries done in beach chair position? **A.** Stroke or even brain death due to ischemia **B.** Only nerve injury **C.** Only hypotension **D.** Only bleeding	**Ans: A**
Q32. Where should the arterial pressure transducer be zeroed for shoulder surgeries in beach chair position? **A.** At the level of the hip **B.** At the level of external auditory meatus to denote cerebral perfusion pressure **C.** At the level of the arm **D.** At the level of the heart	**Ans: B**
Q33. What is cement implantation syndrome (CIS)? **A.** Hypoxia, profound hypotension or even cardiac arrest occurring during cemented prosthesis in total hip replacement **B.** Allergic reaction to cement **C.** Infection due to cement **D.** Local tissue necrosis	**Ans: A**
Q34. What causes cement implantation syndrome? **A.** Blood loss **B.** Infection **C.** Allergic reaction only **D.** Embolization of bone marrow debris, microemboli or fat due to increased intramedullary pressure or reaction to cement (methyl methacrylate)	**Ans: D**
Q35. What is the treatment for cement implantation syndrome? **A.** Oxygenation, adequate hydration and adrenaline for severe reactions **B.** Antihistamines **C.** Only steroids **D.** Only IV fluids	**Ans: A**
Q36. Why is regional anesthesia always preferred over general anesthesia for orthopedic surgeries? **A.** Due to decreased incidence of thromboembolism, decreased blood loss, fewer respiratory complications and better control of pain **B.** Only because of cost **C.** Only because of convenience **D.** Only because of faster recovery	**Ans: A**

Q37. What type of regional block is used for upper limb surgeries?

- **A.** Femoral nerve block
- **B.** Brachial plexus block
- **C.** Epidural
- **D.** Spinal block

Ans: B

Q38. What block can be used for small procedures lasting less than 30-45 minutes?

- **A.** Brachial plexus block
- **B.** Bier's block (IV regional anesthesia)
- **C.** Spinal anesthesia
- **D.** General anesthesia

Ans: B

Q39. What is the most preferred technique for lower limb surgeries?

- **A.** General anesthesia
- **B.** Bier's block
- **C.** Spinal/epidural or combined spinal epidural (most preferred technique)
- **D.** Femoral nerve block alone

Ans: C

Q40. When is general anesthesia required for orthopedic surgery?

- **A.** Only for hip replacement
- **B.** For all surgeries
- **C.** Only for spine surgery
- **D.** For very prolonged surgeries of upper limb, shoulder or spine surgeries or when regional is not possible

Ans: D

Q41. What nerve blocks are frequently utilized for postoperative analgesia in orthopedics?

- **A.** Only local infiltration
- **B.** Only femoral nerve block
- **C.** Continuous adductor canal block for knee, fascia iliaca/femoral nerve block for hip surgeries, brachial plexus block for upper limb
- **D.** Only epidural

Ans: C

Q42. In what position is spine surgery performed?

- **A.** Supine
- **B.** Lateral
- **C.** Sitting
- **D.** Prone position under general anesthesia

Ans: D

Q43. What are the complications related to prone position in spine surgery?

- **A.** Endotracheal tube dislodgement, airway edema, injury to brachial plexus and peroneal nerve, injury to genitalia and breast, ocular injury (postoperative vision loss), hypotension
- **B.** Only hypotension
- **C.** Only nerve injury
- **D.** Only airway problems

Ans: A

Q44. What monitoring is required for major spine surgeries like scoliosis? **A.** Monitoring of spinal cord function by evoked responses like SSEP, MEP and EMG **B.** Only routine monitoring **C.** Only blood pressure **D.** Only pulse oximetry	**Ans: A**
Q45. Is wake up test done in current day practice for spinal cord monitoring? **A.** No, it is not performed in current day practice **B.** Yes, it is the standard **C.** Only in pediatric patients **D.** Only in cervical spine surgery	**Ans: A**
Q46. What should be the concentration of inhalational agents during evoked response monitoring? **A.** Half minimum alveolar concentration (MAC) as inhalational agents can depress evoked responses **B.** Full MAC **C.** 1.5 MAC **D.** No inhalational agents at all	**Ans: A**
Q47. Do IV agents preserve evoked responses? **A.** Only opioids preserve them **B.** No, they suppress them completely **C.** Only ketamine preserves them **D.** Yes, IV agents preserve evoked responses except a little inhibition seen with Propofol	**Ans: D**
Q48. Can muscle relaxants be used during MEP monitoring? **A.** Only non-depolarizing agents **B.** Only succinylcholine **C.** Yes, all muscle relaxants can be used **D.** No, muscle relaxants cannot be used during MEP monitoring	**Ans: D**
Q49. What can happen at the time of tourniquet deflation that is most dangerous? **A.** Only hypothermia **B.** Cardiac arrest **C.** Only pain **D.** Only hypotension	**Ans: B**
Q50. What coagulation changes are seen in fat embolism? **A.** Decreased clotting time **B.** Only increased platelets **C.** Increased clotting time and decreased platelets **D.** No coagulation changes	**Ans: C**

Answer Key & Explanations

Q1. Answer: C — Lengthy procedures, increased blood loss, positioning, tourniquet-related problems, fat embolism and DVT

Orthopedic anesthesia poses special problems due to lengthy procedures, increase blood loss, positioning, tourniquet-related problems, complications like fat embolism and deep vein thrombosis.

Topic: General

Q2. Answer: B — Hypertension, increased cardiac output and increased pulmonary artery pressure

During inflation: Hypertension, increased cardiac output and increased pulmonary artery pressure. This is due to increased blood volume associated with exsanguination.

Topic: Tourniquet Problems

Q3. Answer: A — Local tissue ischemia leads to acidosis (mitochondrial pO_2 becomes zero)

Metabolic changes: Local tissue ischemia leads to acidosis (mitochondrial pO_2 becomes zero within 6 minutes of tourniquet inflation).

Topic: Tourniquet Problems

Q4. Answer: A — Only by deflation (not relieved by IV analgesics)

Pain is due to cellular acidosis, therefore may not be relieved by intravenous analgesics. It is relieved only by deflation.

Topic: Tourniquet Problems

Q5. Answer: B — Sudden release of accumulated metabolites like thromboxane and substance P

Hypotension because of sudden release of accumulated metabolites like thromboxane and substance P.

Topic: Tourniquet Problems

Q6. Answer: A — Due to heat loss as acid metabolites are vasodilators

Hypothermia due to heat loss as acid metabolites are vasodilators.

Topic: Tourniquet Problems

Q7. Answer: D — Transient increase (CO_2 accumulated in limb is released)

Increase in end tidal CO_2 (which is accumulated in limb).

Topic: Tourniquet Problems

Q8. Answer: D — Deflation

Deflation is the most crucial period. Cardiac arrest may occur at the time of deflation due to hypotension and release of acid metabolites.

Topic: Tourniquet Problems

Q9. Answer: B — Not more than 2 hours for both upper and lower limbs

Pressure in tourniquet should not exceed more than 2 hours for both upper and lower limbs.

Topic: Tourniquet Problems

Q10. Answer: D — At least 30 minutes

If re-inflation is needed, then it should be deflated for at least 30 minutes before re-inflating.

Topic: Tourniquet Problems

Q11. Answer: B — Not more than 100 mm Hg above systolic (maximum 250 mm Hg)
Pressure in tourniquet should not exceed more than 100 mm Hg above systolic (maximum-250 mm Hg).

Topic: Tourniquet Problems

Q12. Answer: C — Neuropraxia
Post-tourniquet: Neuropraxia may occur if tourniquet pressure is high or time is prolonged.

Topic: Tourniquet Problems

Q13. Answer: C — After fracture of long bones and pelvis
Usually seen after fracture of long bones and pelvis. Incidence of significant embolism after long bone and pelvic fractures may be as high as 15%.

Topic: Fat Embolism

Q14. Answer: C — Within 24-72 hours
Fat embolism classically presents within 24-72 hours of fracture.

Topic: Fat Embolism

Q15. Answer: B — Higher in nailing compared to plating
Due to rimming the incidence of fat embolism syndrome (FES), is higher in nailing as compared to plating.

Topic: Fat Embolism

Q16. Answer: C — Dyspnea, confusion and petechial hemorrhages
The triad of fat embolism syndrome is dyspnea, confusion and petechial hemorrhages.

Topic: Fat Embolism

Q17. Answer: D — Petechial rash
Petechial rash is pathognomonic of FES.

Topic: Fat Embolism

Q18. Answer: B — Dyspnea, chest discomfort, tachycardia, hypotension, mental clouding, restlessness, seizures, delirium, petechial hemorrhage on upper chest, axilla and conjunctiva, hypoxia and finally coma and death
Signs include dyspnea, chest discomfort, tachycardia, hypotension, mental clouding, restlessness, seizures, delirium, petechial hemorrhage on upper chest, axilla and conjunctiva, hypoxia and finally coma and death.

Topic: Fat Embolism

Q19. Answer: C — Helpful in diagnosis but may not necessarily indicate progression to FES
The fat globules in urine are helpful in diagnosis, but may not necessarily indicate the progression to fat embolism syndrome.

Topic: Fat Embolism

Q20. Answer: C — Diffuse interstitial infiltrates
X-ray chest shows diffuse interstitial infiltrates.

Topic: Fat Embolism

Q21. Answer: D — Fall in end tidal CO2 (ETCO2 may become zero if embolus is large enough to block main pulmonary artery)

Capnography shows fall in end tidal CO_2 ($ETCO_2$ may become zero if embolus is large enough to block the main pulmonary artery).

Topic: Fat Embolism

Q22. Answer: B — Right ventricular strain

ECG shows the right ventricular strain.

Topic: Fat Embolism

Q23. Answer: A — Increased (but has no relation to severity of disease)

Serum lipase is increased (but has no relation to the severity of disease).

Topic: Fat Embolism

Q24. Answer: C — Early stabilization of fracture (may prevent fat embolism)

Stabilization of fracture: Early stabilization of fracture may prevent fat embolism.

Topic: Fat Embolism

Q25. Answer: D — Intermittent positive pressure ventilation (IPPV) with positive end expiratory pressure (PEEP)

If hypoxia persists on oxygen therapy consider intermittent positive pressure ventilation (IPPV) with positive end expiratory pressure (PEEP).

Topic: Fat Embolism

Q26. Answer: B — Fat emboli cause endothelial injury which can precipitate ARDS and cerebral edema

Fat emboli released from long bone causes endothelial injury in the lungs and brain, which can precipitate acute respiratory distress syndrome (ARDS) and cerebral edema respectively.

Topic: Fat Embolism

Q27. Answer: B — No, they are now considered as outdated treatment modalities

Corticosteroids, heparin (which stimulates circulating lipase that breaks neutral fat), ethanol and hypertonic saline are now considered as outdated treatment modalities.

Topic: Fat Embolism

Q28. Answer: C — 1-3%

The incidence of fatal PE after hip surgery can be as high as 1-3%.

Topic: DVT and PE

Q29. Answer: D — Hip surgery patients particularly after prolonged immobilization

The incidence of deep vein thrombosis (DVT) and pulmonary embolism (PE) is highest in orthopedic patients particularly after hip surgery due to prolonged immobilization.

Topic: DVT and PE

Q30. Answer: D — 12 hours or more in preoperative period and restart 12 hours after surgery

Many centers start low molecular weight heparin before 12 hours or more in preoperative period and restart 12 hours after surgery.

Topic: DVT and PE

Q31. Answer: A — Stroke or even brain death due to ischemia

There have been case reports of stroke or even brain death in shoulder surgeries done in beach chair position due to ischemia.

Topic: Positioning

Q32. Answer: B — At the level of external auditory meatus to denote cerebral perfusion pressure

Invasive blood pressure is highly recommended for shoulder surgeries done in the beach chair position with transducer zeroed at the level of external auditory meatus to denote cerebral perfusion pressure.

Topic: Positioning

Q33. Answer: A — Hypoxia, profound hypotension or even cardiac arrest occurring during cemented prosthesis in total hip replacement

Cement implantation syndrome (CIS) occurs more commonly in total hip replacement where cemented prosthesis is used. There occurs hypoxia, profound hypotension or even cardiac arrest.

Topic: Cement Implantation

Q34. Answer: D — Embolization of bone marrow debris, microemboli or fat due to increased intramedullary pressure or reaction to cement (methyl methacrylate)

It may occur because of embolization of bone marrow debris, microemboli or fat due to increased intramedullary pressure created by cement or due to reaction to cement (methyl methacrylate).

Topic: Cement Implantation

Q35. Answer: A — Oxygenation, adequate hydration and adrenaline for severe reactions

The treatment includes Oxygenation, adequate hydration and adrenaline for severe reactions.

Topic: Cement Implantation

Q36. Answer: A — Due to decreased incidence of thromboembolism, decreased blood loss, fewer respiratory complications and better control of pain

Regional anesthesia is always preferred over general anesthesia due to decrease incidence of thromboembolism, decrease blood loss, respiratory complications and better control of pain.

Topic: Choice of Anesthesia

Q37. Answer: B — Brachial plexus block

Upper limb surgeries can be performed under brachial plexus block.

Topic: Choice of Anesthesia

Q38. Answer: B — Bier's block (IV regional anesthesia)

Small procedures lasting less than 30-45 minutes can be performed under Bier's block.

Topic: Choice of Anesthesia

Q39. Answer: C — Spinal/epidural or combined spinal epidural (most preferred technique)

Lower limb surgeries are performed under spinal/epidural or combined spinal epidural (most preferred technique).

Topic: Choice of Anesthesia

Q40. Answer: D — For very prolonged surgeries of upper limb, shoulder or spine surgeries or when regional is not possible

General anesthesia is required either for very prolonged surgeries of upper limb, shoulder or spine surgeries or when regional is not possible.

Topic: Choice of Anesthesia

Q41. Answer: C — Continuous adductor canal block for knee, fascia iliaca/femoral nerve block for hip surgeries, brachial plexus block for upper limb

Continuous adductor canal block for knee replacement, fascia iliaca/femoral nerve block for hip surgeries, brachial plexus block for upper limb surgeries are frequently utilized for postoperative analgesia.

Topic: Pain Management

Q42. Answer: D — Prone position under general anesthesia

Spine surgery is performed under general anesthesia in prone position.

Topic: Spine Surgery

Q43. Answer: A — Endotracheal tube dislodgement, airway edema, injury to brachial plexus and peroneal nerve, injury to genitalia and breast, ocular injury (postoperative vision loss), hypotension

Complications related to prone position include endotracheal tube dislodgement, airway edema, injury to brachial plexus and peroneal nerve, injury to genitalia and breast, ocular injury (post-operative vision loss), hypotension.

Topic: Spine Surgery

Q44. Answer: A — Monitoring of spinal cord function by evoked responses like SSEP, MEP and EMG

Monitoring of spinal cord function is required for major spine surgeries like scoliosis. The function of the spinal cord is monitored by evoked responses like somatosensory evoked potential (SSEP), motor evoked potential (MEP) and electromyogram (EMG).

Topic: Spine Surgery

Q45. Answer: A — No, it is not performed in current day practice

Wake up test done in the past is not performed in current day practice.

Topic: Spine Surgery

Q46. Answer: A — Half minimum alveolar concentration (MAC) as inhalational agents can depress evoked responses

As inhalational agents can depress these evoked responses their concentration should be kept at half minimum alveolar concentration (MAC).

Topic: Spine Surgery

Q47. Answer: D — Yes, IV agents preserve evoked responses except a little inhibition seen with Propofol

IV agents preserve evoked responses except a little inhibition seen with Propofol.

Topic: Spine Surgery

Q48. Answer: D — No, muscle relaxants cannot be used during MEP monitoring

Muscle relaxants cannot be used during MEP monitoring.

Topic: Spine Surgery

Q49. Answer: B — Cardiac arrest

Cardiac arrest can occur at the time of tourniquet deflation.

Topic: Tourniquet Problems

Q50. Answer: C — Increased clotting time and decreased platelets

Coagulation abnormalities like increased clotting time and decreased platelets are seen in fat embolism.

Topic: Fat Embolism